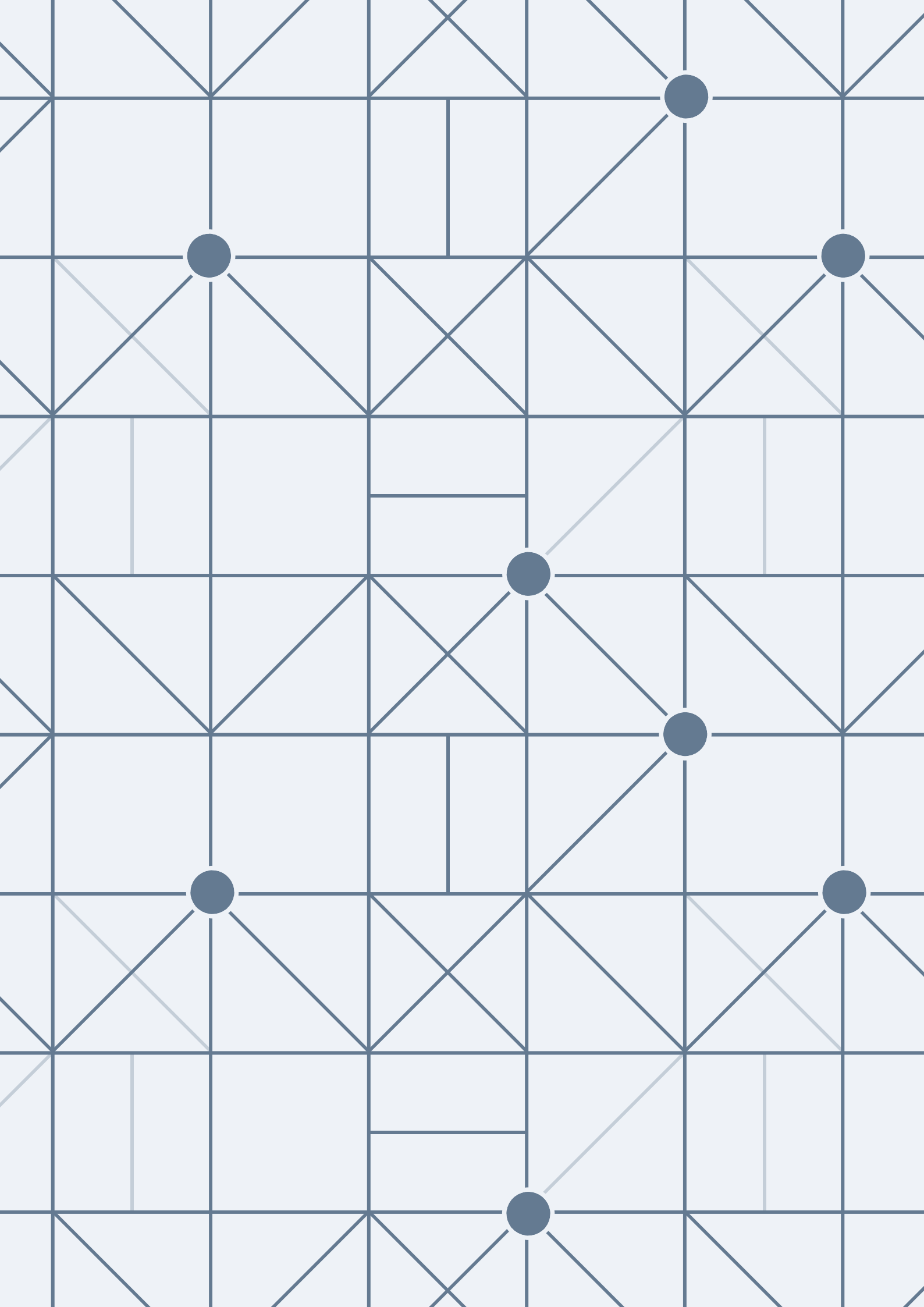




$$A \equiv -\frac{1}{\beta} \ln Z \longrightarrow A = A(N, V, T) = U - TS \quad \text{Helmholtz Free Energy}$$

$$S = -\left. \frac{\partial A}{\partial T} \right|_{N, V} = k_B \left. \frac{\partial \ln Z}{\partial T} \right|_{N, V} = k_B \ln Z$$

$$U = \frac{1}{Z} \left(-\frac{\partial Z}{\partial \beta} \right)_{N, V} = \left(-\frac{\partial \ln Z}{\partial \beta} \right)_{N, V}$$

$$C_V = \left. \frac{\partial U}{\partial T} \right|_{N, V} = k_B \beta^2 \left. \frac{\partial^2 \ln Z}{\partial \beta^2} \right|_{N, V}$$

$$X = \int_0^1 h(x) dx \quad \text{as } X \text{ and } h \text{ are continuous functions}$$

$$X = X(\beta, h(x))$$

$$d(X) = \left. \frac{\partial \ln Z}{\partial \beta} \right|_{h(x)} = \int_0^1 \left. \frac{\partial h(x)}{\partial \beta} \right|_{h(x)} dx$$

$$\left. \frac{\partial \ln Z}{\partial \beta} \right|_{h(x)} = -\left. \frac{\partial \langle h(x) \rangle}{\partial \beta} \right|_{h(x)} = -U$$

$$\left. \frac{\partial \ln Z}{\partial \beta} \right|_{h(x)} = -\left. \frac{\partial \langle h(x) \rangle}{\partial \beta} \right|_{h(x)} = -\frac{1}{Z} \left. \frac{\partial Z}{\partial \beta} \right|_{h(x)} = -\frac{1}{Z} \left(\sum_i e^{-\beta h(x_i)} \right) = -\frac{1}{Z} e^{-\beta h(x)} = \beta h(x) = \beta U$$

$$\int_0^1 h(x) dx = \beta U \Rightarrow X = \beta U \Rightarrow X = U \quad \text{stable for energy}$$

$$\left. \frac{\partial \ln Z}{\partial \beta} \right|_{h(x)} = -U \quad \left. \frac{\partial \ln Z}{\partial \beta} \right|_{h(x)} = -U \quad \left. \frac{\partial \ln Z}{\partial \beta} \right|_{h(x)} = -U$$

